Effects of Spatiotemporal Parameters on Forearm Vibrotactile Stimulus Identification

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***Abstract*—Touch provides a direct and private channel for transmitting information through the skin, offering a foundation for wearable tactile communication. This study examines how spatial and temporal parameters jointly determine the perceptual identifiability and information transfer of vibrotactile stimuli on the forearm. In six experiments, we characterize the effects of actuator density, vibration duration, inter-stimulus interval, and sequence length using multi-actuator arrays. Results showed that localization remained reliable up to four tactors per forearm band but deteriorated with higher spatial density. Shorter vibration durations around 0.5 s preserved accuracy while improving temporal efficiency, and an inter-stimulus interval around 0.5 s effectively reduced sequential interference. Reducing spatial density also mitigated temporal overlap, highlighting an interaction between spatial compactness and temporal clarity. These findings delineate how spatial layout and temporal timing influence identifiability and information transfer on the forearm, and inform the design of compact, efficient wearable tactile displays.**

***Index Terms*—vibrotactile stimuli, identification, forearm, wearable, array**

I. Introduction

Human communication relies predominantly on vision and audition, yet both channels can become overloaded or inaccessible in certain situations. Touch offers a direct and private alternative that can convey information through the skin without competing with other sensory modalities. This modality has long been explored for communication purposes, from early sensory substitution methods to modern vibrotactile systems [1], [2]. In parallel, progress in on-body tactile communication has been driven by the use of wearable technologies [3], [4].

Wearable tactile displays deliver information in the background while other sensory channels are in use [5]–[11]. In this context, the forearm has emerged as a promising site for wearable vibrotactile displays given its usable surface and adequate usability [12]–[15]. Building on this platform, recent systems have moved beyond alert-based feedback to structured communication, achieving reliable phoneme- and word-level recognition through the skin [16], [17].

Effective tactile communication requires considering both spatial and temporal factors for stimulus identification. For example, small inter-tactor distances or brief inter-stimulus intervals in multi-stimulus sequences can induce spatiotemporal interference. When multiple tactors are activated in proximity or in short succession, interference across both dimensions can reduce discriminability and limit the information bandwidth through the skin. A systematic understanding of this spatiotemporal interaction is therefore fundamental to designing expressive and reliable tactile codes.

The present study seeks to establish such a perceptual foundation for the forearm as a site for tactile communication. We conducted six experiments to quantify how spatial and temporal parameters jointly affect tactile identifiability and information transfer. Experiment (Exp.) 1, 2, and 3 examined spatial factors, including the numbers and arrangements of tactors across forearm bands (Sec. IV). Exp. 4, 5, and 6 investigated temporal factors, such as vibration duration, inter-stimulus interval, and sequence length, to assess temporal integration and masking (Sec. V). Across all conditions, we analyzed accuracy, confusion structures, and information transfer to derive design guidelines for effective and efficient spatiotemporal information encoding.

Our long-term goal is to develop tactile communication systems that can convey linguistic and symbolic information through the skin, with a particular focus on the Korean language. To this end, this work provides a comprehensive empirical foundation by identifying the perceptual limits and optimal ranges of spatiotemporal parameters for vibrotactile stimuli and patterns applied to the forearm.

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II. Related Work

A. Tactile Communication

Early tactile communication methods, such as the Tadoma technique and electrotactile hearing aids, demonstrated that linguistic information could be conveyed through the skin [18], [19]. Although these systems enabled basic speech perception, their performance was limited by actuator precision and strong perceptual interference across stimulation sites [20]–[22].

Advances in miniature actuators and digital control have made vibrotactile communication a practical and scalable alternative. Recent studies showed that structured vibration patterns can represent linguistic or symbolic units, enabling users to recognize English phonemes with a mean accuracy of 85% and even words with accuracies of 87% on a trained set and 77% on a novel set. [23], [24]. Such results suggest that touch-based communication can extend beyond alerts or cues to support symbolic and linguistic expression. Furthermore, vibrotactile stimulation can be favored for its technical simplicity, energy efficiency, and high social acceptability compared with electrical or pressure-based methods [25], [26].

B. Wearable Vibrotactile Displays

Vibrotactile communication has been integrated into wearable systems designed for navigation, affective feedback, and situational awareness [9], [27]–[29]. Wearable vibrotactile displays have evolved with advances in actuator, battery, and wireless technologies [30]. These systems embed small stimulators on the body to deliver structured tactile cues in a compact and unobtrusive form [5], [6], [8]–[10]. These developments enable wearable displays that deliver tactile information for both assistive and general communication purposes, marking a shift toward compact, multimodal interfaces that can be seamlessly incorporated into everyday life.

Research has explored a variety of body sites, each offering distinct trade-offs between perceptual resolution and wearability [8], [16], [31], [32]. Researchers have successfully implemented systems by strategically arranging tactors to exploit the perceptual characteristics of each region. For instance, torso- or waist-mounted arrays were often designed with a grid layout to provide a broad spatial range for directional guidance [32]–[36], leveraging the body's natural sense of orientation. Conversely, systems designed for the arms and wrists focus on compact, band-like forms to offer greater convenience for everyday use. For instance, tactile displays on the arm have been implemented by arranging tactors in circumferential and longitudinal lines to investigate how multi-tactor placement affects localization performance [13], [37]–[41]. Among these, the forearm offers a favorable compromise between acuity and wearability [12], [42], [43].

C. Forearm-Mounted Vibrotactile Displays

Prior work has demonstrated robust localization ability [8], [12], [44] and successful linguistic encoding through forearm-mounted vibrotactile display [16]. For example, phoneme-based vibrotactile communication on a 4×6 array achieved accuracy above 80% [17]. However, dense or simultaneous activation leads to pronounced degradation, with localization performance dropping below 20% [43]. This decline indicates that mechanical coupling and temporal overlap jointly constrain the effective information bandwidth of the forearm. While some studies investigated optimal stimulus durations between 100–400 ms [45], few have analyzed the combined influence of spatial and temporal factors [46], [47].

D. Design Factors for Vibrotactile Displays

The perceptual capacity of vibrotactile displays depends critically on the interplay between spatial and temporal design factors. Spatial parameters, such as actuator spacing and number, determine localization precision and potential inter-site interference [43], [44], [48]. Increasing the number of tactors can enrich representational capacity but often reduces discriminability; accuracy has been shown to decrease from 84% in a 3×3 array to 59% in a 4×4 configuration [49], [50].

Temporal parameters govern the perceptual integration of sequential stimuli [51], [52]. Sequential presentation consistently outperforms simultaneous activation, with accuracy reaching 93.2% compared to 26.2% in a one-time presentation [52]. Effective stimulus durations typically range between 30 and 400 ms, depending on frequency and intensity [45], [53]. Other perceptual dimensions, such as intensity, frequency, and duration, also modulate subjective impressions of roughness or strength [54].

Most prior work on tactile communication has treated spatial and temporal factors separately, leaving their combined effects on perceptual performance and information transfer underexplored. Establishing this link is a key to wearable systems that are both expressive and robust. We address this gap by systematically testing spatiotemporal interactions on the forearm-based tactile communication and identifying settings that improve identifiability and information transfer.

III. General Methods

This paper investigates the effects of spatiotemporal factors on the identification of vibrotactile stimuli applied to the forearm. We conducted six experiments, with the first three (Exp. 1, 2, and 3) for spatial parameters such as the number and arrangement of tactors and the rest three (Exp. 4, 5, and 6) for temporal parameters including duration and inter-stimulus interval (ISI). Rather than exhaustively testing all parameters at once, we incrementally expanded the design space, using simpler spatial layouts or temporally isolated stimuli to establish baselines and subsequently probing denser configurations or tighter temporal spacing. This stepwise approach resulted in partial overlap between experiments, but was necessary to identify perceptual boundaries in a systematic manner. All experimental procedures were approved by the Institutional Review Board at POSTECH (No. PIRB-2025-E010). This section describes the apparatus, stimuli, and participants common to all six experiments.

A. Apparatus

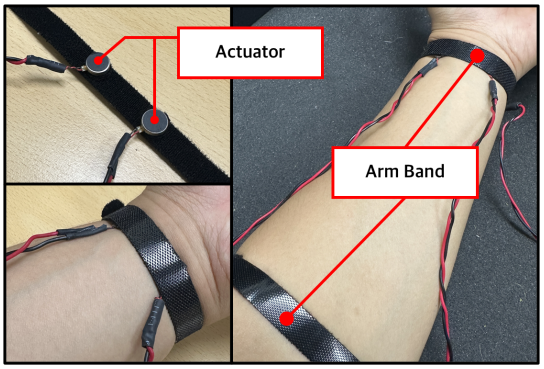

Fig. 1. Experimental setup. Example setup showing placement of tactors on the distal and proximal regions of the forearm using a velcro band.

We used eccentric rotating mass (ERM) motors (Seeed Technology, 10 mm diameter) as vibrotactile actuators. ERM motors were selected for their mechanical robustness, simple driving circuitry, and suitability for compact, high-density actuator arrays. Each motor was independently driven by a pulse-width modulation (PWM) signal from an Arduino Due.

The tactors were mounted on adjustable velcro bands such that the contact surface of each motor directly touched the skin (Fig. 1). Depending on experimental conditions, tactors were positioned along the distal–proximal axis of the forearm and distributed on the dorsal, palmar, medial, and lateral surfaces. To accommodate inter-individual differences in forearm size, tactor positions on each band were adjusted in proportion to the forearm circumference of a participant, while preserving the relative spatial configuration across participants. Circumferences at the distal and proximal forearm were measured for each participant and used to scale the bands accordingly. Longitudinal inter-band spacing was also determined in proportion to the forearm length between the distal and proximal regions, ensuring consistent relative spacing across participants. Each band was adjusted to provide firm but comfortable contact with the skin, also accommodating individual differences in arm circumference. Participants placed their forearm on a padded support, viewed a monitor showing a graphical user interface (GUI), and wore noise-canceling headphones to eliminate auditory cues.

B. Stimuli

Each stimulus consisted of a single mechanical vibration generated by one ERM motor at full drive intensity. The vibration frequency was approximately 160 Hz with a peak acceleration of 1 G. Other temporal parameters of stimuli are specified for each experiment later.

C. Participants

Ten participants participated in each of the six experiments. Demographic details for each experiment are summarized in Table I. All participants were students enrolled in the authors' university. Each participant took part in only one experiment, and all participants were naive to the specific experimental condition in which they participated. All participants reported normal tactile sensitivity and no history of sensory disorders. They provided informed consent and were compensated for their participation. Each participant received approximately USD 14 upon completion of the experiment.

TABLE I
PARTICIPANT DEMOGRAPHICS.

Exp.	Number	Male	Female	Age (M)	Age (SD)
1	10	6	4	25.10	2.42
2	10	6	4	24.80	4.10
3	10	6	4	24.20	1.93
4	10	6	4	26.20	4.21
5	10	7	3	27.60	3.03
6	10	4	6	23.50	3.69

IV. Exp. 1, 2, and 3: Effects of Spatial Factors

These three experiments examined how the spatial configuration of vibrotactile arrays on the forearm influences identification performance and information transfer while systematically varying the number of tactors within each band and the number of bands across the arm. The goal was to find the spatial conditions that allow reliable identification of multiple tactors while maintaining compact array layouts. These spatial layouts would serve as a reference for the temporal factor investigations described later in Sec. V.

A. Methods

1) Experimental Conditions: All spatial configurations used in Exp. 1–3 are illustrated in Fig. 2. Each diagram shows cross-sectional layouts of tactors mounted on different forearm bands (D = distal, P = proximal, C = central, SD = sub-distal, SP = sub-proximal), arranged along palmar, dorsal, medial, and lateral sides. Within each configuration, all arm bands shared the same tactor arrangement.

Exp. 1 examined the basic localization with a small to moderate number of tactors. Tactors were placed on the distal and proximal bands, with four, six, or eight actuators distributed across the four anatomical sides (Fig. 2b, top row). Exp. 2 maintained the same distal-proximal bands but increased the number of tactors per band. As shown in the middle row of Fig. 2b, palmar and dorsal sides were expanded to two or three tactors, yielding denser layouts up to 16 tactors. This design allowed us to probe the perceptual limit of identifying multiple tactors within the same anatomical region. Exp. 3 extended the layouts by adding intermediate bands between the distal and proximal regions. One or two additional bands (sub-distal, central, and sub-proximal) were introduced, producing three-band or five-band configurations (Fig. 2b, bottom row). Within these arrays, tactors were distributed across palmar, dorsal, medial, and lateral sides, and the number of tactors per band was systematically varied.

All vibration stimuli has a fixed duration of 2 s in Exp. 1–3, which yielded stable perceptual cues to focus on the analysis of spatial factors.

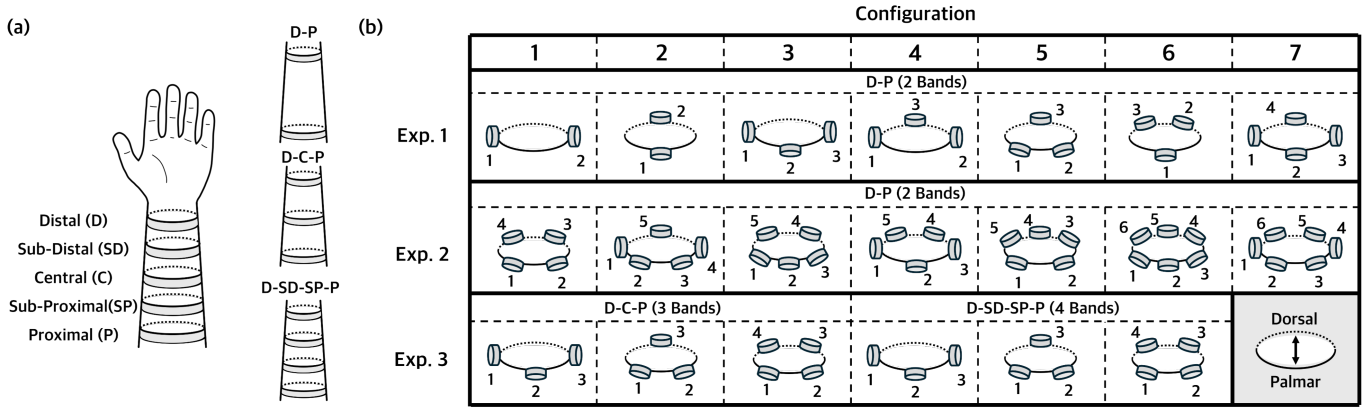

Fig. 2. Spatial configurations of tactors used in Exp. 1–3. (a) Forearm regions and anatomical sides. Five horizontal arm bands are defined: distal (D), sub-distal (SD), central (C), sub-proximal (SP), and proximal (P). (b) Tactor layouts used in experiments and configurations. Each diagram shows a cross-sectional view of the forearm cut along the dorsal-palmar axis, with tactors placed on palmar, dorsal, medial, and lateral sides. Each cell corresponds to one configuration, and the band codes (D–P, D–C–P, and D–SD–C–SP–P) above each group indicate the forearm lines included in that configuration.

2) Task and Procedure: Participants' task was to identify the location of a single vibrotactile stimulus applied to the forearm. Exp. 1, 2, and 3 consisted of seven, seven, and six sessions, respectively, with each session corresponding to a different spatial configuration. The session orders were identical for all participants, progressing from simpler to more complex configurations (e.g., sparser to denser and fewer to more bands). Pilot tests indicated that direct exposure to high-density configurations induced substantial cognitive overload, resulting in unstable perceptual performance. Thus, a progressive ordering was adopted to support task adaptation and to obtain the identifiability limits under stable perceptual conditions. At the beginning of every session, the GUI displayed the tactor array configuration so that participants could familiarize themselves with the layout.

During a trial, participants pressed the *Stimulus* button to trigger the vibration stimulus assigned to the trial. Then, they reported the perceived location by clicking the corresponding icon on the GUI and pressed the *Next* button to proceed to the next trial. This sequence was repeated until all trials for the session were completed.

In every session, each actuator location was stimulated five times in random order. Exp. 1, 2, and 3 included 200, 360, and 350 trials, respectively. At the end of each session, participants took a two-minute break to prevent fatigue and tactile adaptation, and the motor bands were reconfigured for the next spatial configuration.

3) Data Analysis: The identification performance was quantified using two measures: the percent-correct (PC) score and the information transfer (IT). The PC score was defined as the proportion of correct responses out of the total number of trials in percentage.

The IT was estimated from the confusion matrix between presented and reported stimuli, following the standard information-theoretic formulation [26], [55]. Let k be the number of possible stimuli, n the total number of trials, and

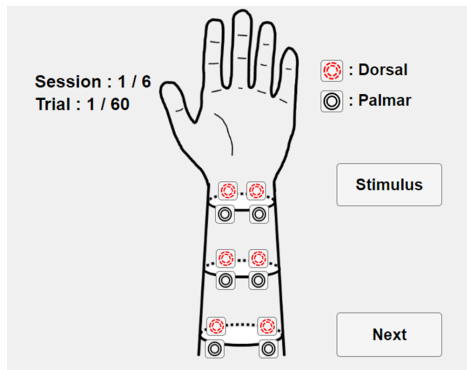

Fig. 3. Graphical user interface (GUI) used for Exp. 1–3. The layout mirrored the physical configuration of tactors on the forearm.

$n_{i,j}$ the number of responses j given to stimulus i. Then,

$$IT_{est} = \sum_{i=1}^{k}\sum_{j=1}^{k}\left(\frac{n_{i,j}}{n}\right)\log_2\frac{n_{i,j}\,n}{n_{i,*}\,n_{*,j}}, \tag{1}$$

where $n_{i,*}$ and $n_{*,j}$ denote the row and column sums of the confusion matrix, respectively.

This measure quantifies the mutual information between presented and reported stimuli, capturing not only the proportion of correct responses but also the confusion patterns. Hence, it reflects the information transmitted through the stimulus-response channel for each configuration. An IT_{est} of b bits corresponds to identifying 2^b stimuli with perfect accuracy. The theoretical upper bound of information transfer, IT_{max}, was determined by $IT_{max} = \log_2 k$.

B. Results

1) Exp. 1 (Two-Band Arrays): Exp. 1 examined how the number of actuators per band affected localization (identification) performance. Participants experienced configurations with two, three, and four tactors arranged in two bands along the forearm. As shown in Fig. 4a, their performance was

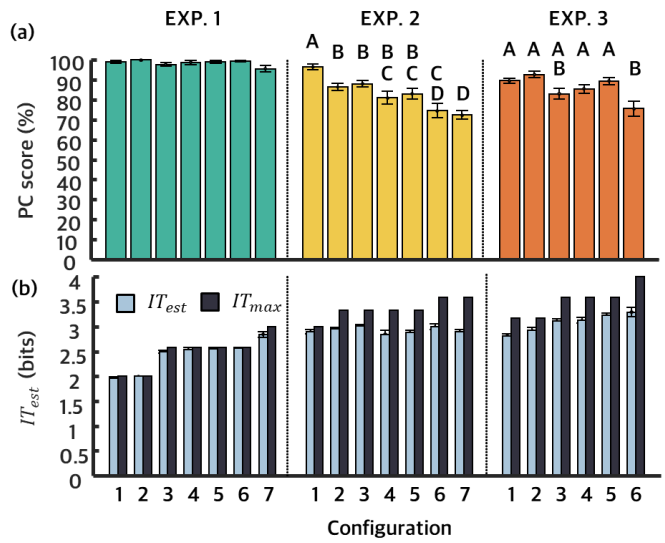

Fig. 4. Results of Exp. 1–3 for the effects of spatial factors. Error bars represent standard errors. (a) PC scores. Conditions marked with different letters had significant differences by Tukey's HSD tests. (b) Estimated information transfer values (IT_{est}) and their theoretical upper bounds (IT_{max}).

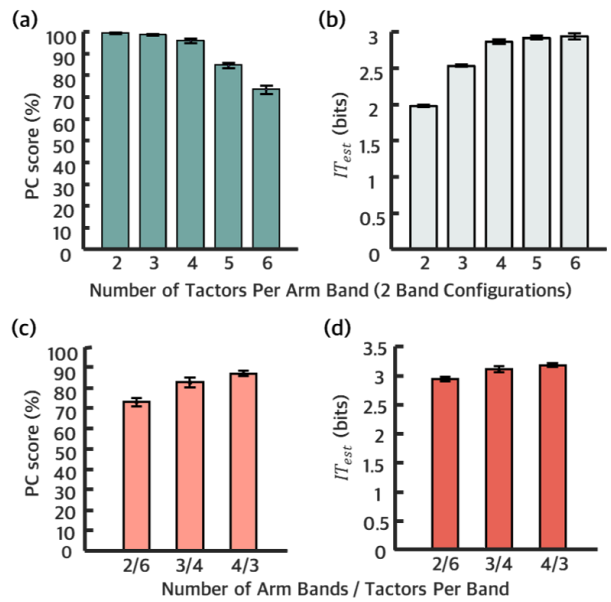

Fig. 5. Identification performance as a function of tactor density or array configuration. (a)–(b) show the PC scores and IT for two-band layouts as the number of tactors per band increases from 2 to 6. (c)–(d) show the performance for configurations with a total of 12 tactors across different numbers of arm bands.

consistently high across all layouts, with mean PC scores exceeding 95%. A slight decline was observed as the spatial density increased, but the reduction was marginal even in the densest configuration.

Information-transfer analysis further supported this observation (Fig. 4b). The estimated information transfer (IT_{est}) closely approached the theoretical upper bound (IT_{max}) across all conditions. For layouts with two or three tactors per band, the difference between IT_{max} and IT_{est} remained below 0.01 bits, indicating nearly perfect identification. Even in the four-tactor condition, where the spatial density was highest, the gap was only 0.15 bits ($IT_{est} = 2.84$ bits and $IT_{max} = 3.00$ bits), confirming that participants retained high identification ability despite the increased proximity between tactors. These results demonstrate that the tactor identification based on its location on the forearm remains robust for up to four tactors per band.

2) Exp. 2 (Effects of Array Density): Exp. 2 examined how increasing the number of tactors affected identification across the seven configurations of growing spatial complexity. As shown in Fig. 4a, the accuracy progressively decreased as the array density increased. The mean PC score was 96.5% in the 8 tactor configuration (Configuration 1) but declined to 84.6% for 10 tactors (Configuration 2–5) and further to 73.4% for 12 tactors (Configuration 6–7). This trend indicates a clear degradation in localization accuracy with increasing tactor density. A repeated-measures ANOVA with *Configuration* as the within-subject factor revealed a significant main effect of tactor configuration on the identification accuracy ($F(6, 54) = 16.98$, $p < .0001$). Post-hoc Tukey HSD tests showed that the identification accuracy decreased significantly from 8, 10, and 12 tactors configurations.

The information-transfer analysis showed a consistent trend (Fig. 4b). Although IT_{est} decreased with the tactor density, the rate of decline was gradual. In the 8-tactor condition, IT_{est} reached 2.89 bits with only a 0.11 bit difference from IT_{max} (3.00). Adding more tactors slightly increased the theoretical capacity (up to $IT_{max} = 3.58$ bits in configurations 7), but the corresponding IT_{est} values remained nearly constant around 2.9 bits, increasing the gap to 0.70 bits in the densest configuration. This indicates that although additional tactors theoretically increase the available information transmission capacity, users cannot effectively utilize the added capacity due to the increased spatial interference.

3) Exp. 3 (Multi-Band Configurations): Exp. 3 investigated how distributing tactors across the multiple forearm bands affects the identification performance. Unlike Exp. 2, where PC scores decreased monotonically with increasing density, here the performance varied primarily with array configuration (Fig. 4a). Most 12-tactor layouts achieved high PC scores between 85% and 93%, while the 16-tactor layout showed a pronounced drop to 76%. A repeated-measures ANOVA with *Configuration* as a within-subject factor confirmed a significant main effect ($F(5, 45) = 7.58$, $p < .0001$). Post-hoc Tukey HSD indicated that the 16-tactor configuration yielded significantly lower accuracy than all 12-tactor configurations (all $p < .01$), whereas differences among the 12-tactor layouts were not significant ($p > .2$).

This observation was also supported by the IT analysis (Fig. 4b). For the 12-tactor configurations, IT_{est} ranged from 3.11 to 3.23 bits, which was 0.5 bits below their theoretical maximum ($IT_{max} = 3.58$ bits). However, for the 16-tactor layout, the difference was increased to 0.71 bits ($IT_{est} = 3.29$ bits and $IT_{max} = 4.00$ bits).

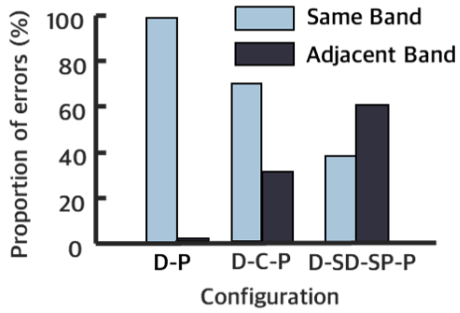

Fig. 6. Distribution of mislocalization errors across arm bands.

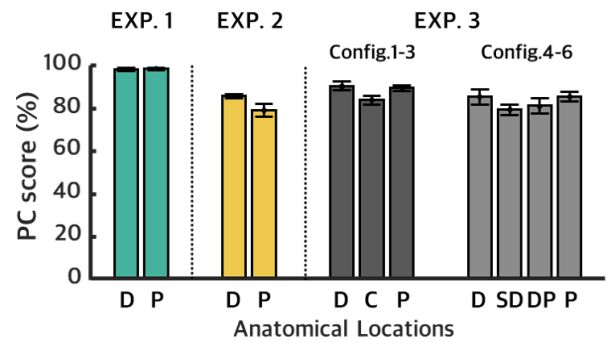

Fig. 7. Percent-correct (PC) scores across anatomical locations of the forearm in Exp. 1–3.

C. Discussion

1) Effects of Tactor Density Per Band: The first observation from the three experiments concerns the perceptual limitation imposed by tactor density along a single arm band. As shown in Fig. 5a–b, PC scores remained consistently high, exceeding 95%, when up to four tactors were arranged along each band. Beyond this level, performance declined steeply—to 85% with five tactors and 73% with six—while IT_{est} increased only marginally from 2.84 to 2.89 bits. This divergence between PC and IT_{est} indicates that although the overall information space expands with added tactors, users cannot effectively distinguish or identify spatially overlapping stimuli. Taken together, four tactors per band is an adequate upper bound for reliable localization under our hardware and task.

2) Effects of Band Count: While increasing tactor density per arm band over four led to perceptual interference, redistributing the same number of tactors across multiple bands improved overall performance. Fig. 5c–d illustrate this trend for 12-tactor arrays with varying numbers of arm bands. The mean PC score increased from 73% in the two-band layout to 83% in the three-band layout and then to 87% in the four-band layout. Similarly, IT_{est} rose from 2.94 bits to 3.10 bits and 3.18 bits, indicating that spreading tactors over a wider skin area enhances perceptual identifiability.

On the other hand, as shown in Fig. 6, increasing the number of bands also altered error patterns. When only two bands were used, almost all errors occurred within the same band. However, with three or four bands, mislocalizations between adjacent bands became more frequent, accounting for about one-third and more than half of the total errors, respectively. Moreover, the proportion of inter-band errors was not correlated with the forearm length of a participant ($R^2 = 0.14$), suggesting that the observed changes in error patterns were primarily driven by layout configuration rather than individual arm size. These patterns demonstrate a clear trade-off: while additional bands improve overall identifiability, they also reduce inter-band spacing, increasing the likelihood of adjacent-band confusions. Therefore, no single configuration can be considered universally optimal; instead, the ideal design should balance spatial coverage and separation. Based on these results, the 12-tactor multi-band layouts, which use three or four tactors per band distributed over three or four arm bands, achieve the best balance between density and coverage. They can provide a robust foundation for investigating how temporal modulation further influences tactile stimulus identification in Exp. 4–6.

3) Effects of Anatomical Locations: Fig. 7 summarizes the identification performance across anatomical locations observed in Exp. 1–3. In Exp. 1, where sparse two-band layouts were used, performance was uniformly high for both distal (D) and proximal (P) bands, with mean PC scores exceeding 98%. This indicates that, under low spatial density, anatomical location had little influence on identification accuracy.

As the spatial complexity increased, location-dependent differences emerged. In Exp. 2, identification accuracy decreased overall, with the distal band (85.5%) exhibiting higher PC scores than the proximal band (78.9%). Similarly, in Exp. 3, performance varied across band locations depending on the array configuration. In the three-band layouts (Configuration 1–3), the distal and proximal bands showed higher accuracy than the central band, whereas in the five-band layouts (Configuration 4–6), the intermediate bands (SD and SP) yielded lower PC scores than the distal and proximal bands.

These results suggest that anatomical location can affect identification performance when spatial layouts become denser or more complex. Under configurations with increased spatial density, the distal band consistently supported accurate identification with a larger number of tactors compared to more proximal or intermediate bands. Also, locations near the distal and proximal ends of the forearm tended to support more reliable identification than intermediate regions. These trends are consistent with prior findings reporting enhanced tactile localization near joint-adjacent areas [13], and may reflect differences in tactile sensitivity or spatial acuity along the forearm.

4) Effects of Anatomical Variation: To ensure that the observed performance trends were not due to anatomical differences among the participants, we examined the relationship between forearm circumference and PC score (Fig. 8). A regression analysis revealed no significant correlation ($R^2 = 0.0217$), suggesting that perceptual performance is largely independent of arm size. Thus, the spatial organization of tactors, rather than individual anatomical variation, primarily

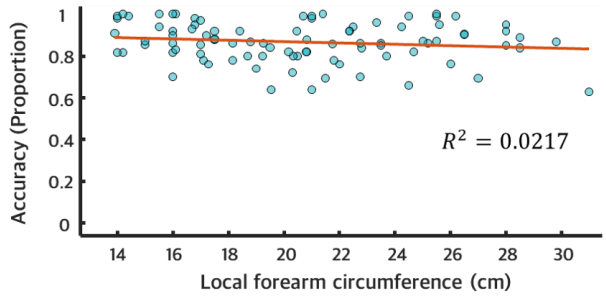

Fig. 8. Localization accuracy for each local forearm band plotted against the corresponding local forearm circumference.

determined the perceptual effectiveness of the array design.

V. Exp. 4, 5, and 6: Effects of Temporal Factors

These three experiments examined how temporal parameters influence the identification of vibrotactile stimuli applied on the forearm. Using the effective spatial configurations found in Sec. IV, we examined how the temporal structure of stimulation—defined by vibration duration, inter-stimulus interval (ISI), and sequence length—affects the identifiability of successive vibrotactile pulses. The goal was to characterize how effectively tactile information can be conveyed over time and determine the temporal conditions that enable reliable sequential perception.

A. Methods

1) Experimental Conditions: The spatiotemporal conditions for Exp. 4–6 are illustrated in Fig. 9. Exp. 4 and Exp. 5 used the 12-tactor multi-band array found best in Sec. IV, with tactors distributed over the palmar, dorsal, medial, and lateral sides of the forearm. Exp. 6 employed a reduced 9-tactor configuration arranged over three arm bands.

Exp. 4 examined how vibration duration affects single-stimulus identification. Three spatial configurations were used, including two layouts with three tactors per band across four arm bands and one layout with four tactors per band across three arm bands. Each configuration was presented with vibration durations of 1.0 s and 0.5 s to determine whether shorter stimuli, which allow faster stimulus delivery, diminish identifiability.

Exp. 5 investigated the effects of temporal interval on identifying two consecutive vibrations. Using the 12-tactor spatial configuration that yielded the best single-stimulus performance in Exp. 4 and a fixed 0.5 s duration, participants identified pairs of stimuli with ISIs of 0, 0.5, and 1.0 s between the first and second vibrations. When ISI was set to 0, the second vibration was triggered immediately after the offset of the first, without any temporal gap. Thus, even when two stimuli were delivered at the same physical location, they were implemented as two successive motor activations. Because the ERM motor was fully turned off between activations, the re-triggering introduced a brief transient that perceptually separated the two

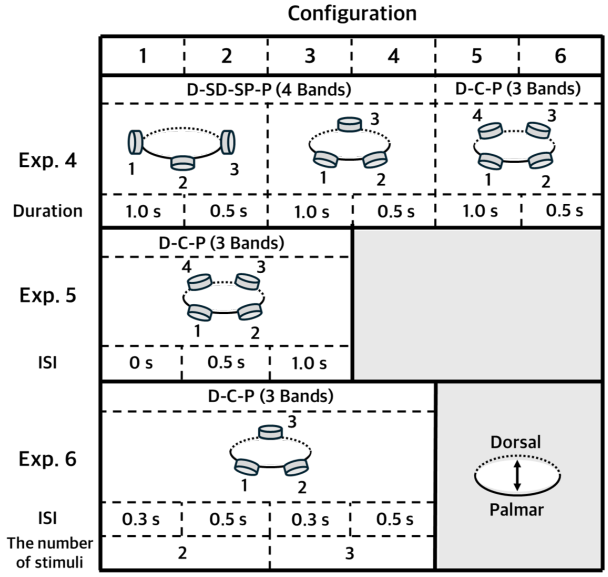

Fig. 9. Spatiotemporal conditions used in Exp. 4–6. Each diagram shows a cross-sectional layout of tactors mounted on the forearm, cut along the dorsal-palmar plane.

pulses. A minimum duration of 0.5 s was used to guarantee that the ERM motor reached full saturation before offset.

Exp. 6 extended the coverage to sequences of two and three vibrations. To mitigate the performance degradation observed with the 12-tactor configuration in Exp. 5 under sequential stimulation, the number of tactors was reduced to nine in Exp. 6. The duration of each vibration was fixed at 0.5 s, which provided single-stimulus performance comparable to 1.0 s in Exp. 4 while improving temporal efficiency. As for ISI, 0.5 s and 0.3 s were used to probe shorter temporal separations, while the 1.0 s interval from Exp. 5 was excluded because no substantial gain was observed beyond 0.5 s. Each three-vibration sequence used the identical intervals between successive vibrations.

For two-vibration sequences (Exp. 5 and Exp. 6), all ordered pairs of the physical tactor locations were used (12^2 for Exp. 5 and 9^2 for Exp. 6), preserving the full spatial resolution of each array. For three-vibration sequences in Exp. 6, however, the full combinatorial space ($9^3 = 729$) would have imposed excessive cognitive demands on participants with an impractically long experimental time. To keep the task tractable while preserving spatial diversity, the sequence set was therefore defined over six abstract spatial sites ($6^3 = 216$).

Specifically, two adjacent palmar tactors on the same band were treated as a single abstract site when generating three-stimulus sequences. During the experiment, physical stimulation was delivered to one of the two palmar tactors, with each selected in equal proportion across trials. This procedure preserved the effective spatial diversity of stimulation while substantially reducing the number of sequence combinations and the associated cognitive load.

2) Task and Procedure: In Exp. 4, participants reported the perceived location of a single vibration. In Exp. 5 and 6, stimuli consisted of two or three consecutive vibrations, and participants were instructed to identify their locations and presentation order.

In Exp. 5 and 6, participants went through a brief practice session before the main session. Each tactor location was stimulated five times in random order so that participants would experience vibration sequences stimulating different locations.

In each trial, participants pressed the "Stimulus" button on a GUI to trigger the stimulus and reported perceived locations by clicking the corresponding icons in the perceived order (Fig. 10). In Exp. 5 and 6, each click generated a numbered marker (1, 2, $\cdots$) representing the perceived order. After entering responses, participants pressed the "Next" button to move on to the next trial. Between sessions, a two-minute rest was provided to prevent fatigue and tactile adaptation.

Exp. 4, 5, and 6 included 360 trials (single vibration, 12 locations $\times$ 6 sessions $\times$ 5 repetitions), 432 trials (12 locations $\times$ 12 locations $\times$ 3 sessions), and 594 trials (two vibrations: 9 locations $\times$ 9 locations $\times$ 2 sessions and three vibrations: 6 locations $\times$ 6 locations $\times$ 6 locations $\times$ 2 sessions), respectively. Trial orders were randomized within each session.

3) Data Analysis: In addition to PC scores, information transfer (IT) was used as the primary quantitative metric of perceptual performance. In Exp. 4, where a single vibration was presented ($m = 1$), the information transfer was estimated using (1) as in Exp. 1–3.

For Exp. 5 and 6, which involved sequential presentations of multiple vibrations, both IT_{est} and IT_{max} were extended to account for the temporal additivity across m successive vibrations. Following Durlach's additivity principle [56], the total information transmitted by a sequence of m stimuli $(S_1, S_2, \cdots, S_m)$ was defined as the sum of the information transfers contributed by individual stimuli:

$$IT_{\text{est}} = \sum_{i=1}^{m} IT(S_i). \tag{2}$$

This approximation interprets each presentation in the sequence through an independent channel, yielding an additive estimate of total information.

B. Results

1) Exp. 4 (Effects of Stimulus Duration): Exp. 4 examined how vibration duration influences the identification of single vibrotactile stimuli. Three spatial configurations were reused from Exp. 3 (Fig. 9). Two layouts placed three tactors per band across four arm bands, and one layout placed four tactors per band across three arm bands. Each configuration was tested with vibration durations of 1.0 s and 0.5 s.

Identification performance was generally consistent across durations within each configuration (Fig. 11a and b). In the four-band layouts, mean PC scores were 79.0% and 75.5% in one layout and 74.8% and 74.0% in the other, whereas the three-band layout yielded noticeably higher scores of 83.2% and 84.3%. The corresponding IT_{est} values ranged from 2.93 to 3.10 bits and followed the same pattern. A repeated-measures ANOVA revealed a significant main effect of spatial configuration ($F = 4.16$, $p = 0.0327$), but no main effect of duration ($F = 0.33$, $p = 0.5799$) or their interaction term ($F = 1.21$, $p = 0.3204$). Post-hoc analyses indicated that the four-tactor-per-band configuration yielded higher identification accuracy than the three-tactor-per-band configurations, while no reliable difference was observed between the two four-band configurations.

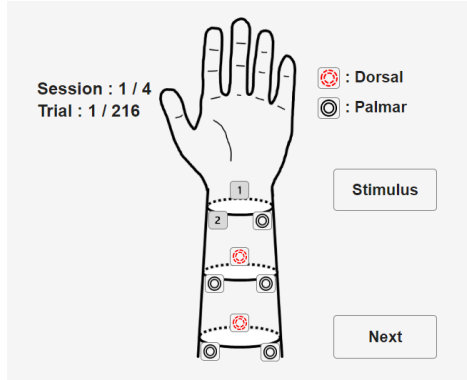

Fig. 10. GUI used in Exp. 4–6. The layout mirrored the physical configuration of tactors on the forearm. Participants pressed the "Stimulus" button to trigger a vibration stimulus. They selected the perceived locations by clicking the corresponding tactor symbols in the perceived order.

Overall, vibration duration, in the tested range, had little influence on single-stimulus identification. Spatial configuration was the primary factor determining the identification performance, and the layout with four tactors per band across three arm bands provided the most reliable localization across durations. This configuration was therefore adopted as the standard for the following temporal experiments.

2) Exp. 5 (Effects of Inter-Stimulus Interval): Exp. 5 investigated how the temporal interval between two consecutive vibrations affects sequential localization. Using the 12-tactor configuration, found as the best in Exp. 4 for single vibrations, and a fixed 0.5-s duration, participants identified two successive vibrations presented with ISIs of 0.0, 0.5, or 1.0 s. The first-vibration PC score averaged 74.1%, while the second-vibration PC score decreased to 59.5% (Fig. 11a). This tendency was consistent across all ISIs. The full-sequence PC score was relatively low: from 44.0% at 0.0 s to 50.6% at 0.5 s and 50.5% at 1.0 s. A repeated-measures ANOVA on full-sequence PC score showed a significant main effect of ISI ($F = 3.77$, $p = 0.0428$). Post-hoc Tukey tests showed that accuracy at 0 s (44.0%) was significantly lower than at 0.5 s (50.6%) and 1.0 s (50.5%) ($p < .05$), with no difference between 0.5 s and 1.0 s ($p = .98$). Consistently, IT_{est} increased with ISI (4.64, 4.87, 4.95 bits at 0, 0.5, 1.0 s, respectively), while IT_{max} remained 7.17 bits (Fig. 11b).

These findings show that moderate temporal spacing around 0.5 s effectively mitigates interference between successive vibrations, whereas longer intervals yield little additional benefit.

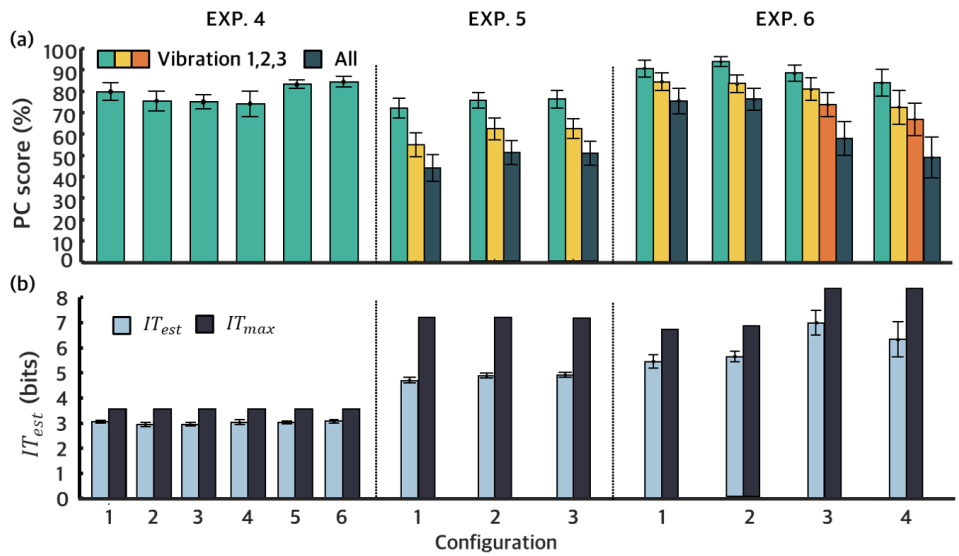

Fig. 11. Results of Exp. 4–6. Error bars indicate standard errors. For sequential stimuli, "Vibration n" denotes the identification accuracy for the n-th vibration in the sequence, and "All" represents the accuracy of correctly identifying all vibrations within a trial.

It is also notable that IT_{est} was considerably smaller than IT_{max} for these two-vibration sequences.

3) Exp. 6 (Effects of Vibration Sequence Length and Inter-Stimulus Interval): Exp. 6 extended the analysis to multi-vibration sequences by comparing two- and three-vibration presentations. The number of tactors was reduced to nine based on the low identification performance of two-vibration sequences observed in Exp. 5. The ISIs (0.3 s and 0.5 s) and the 0.5-s duration were adopted to evaluate the lower temporal bound for effective sequential perception.

Across ISIs, identification performance declined with increasing sequence length. Full-sequence PC score averaged 72.4% for two-vibration sequences and 50.9% for three-vibration sequences. A two-way repeated-measures ANOVA confirmed the significance of the two main effects, sequence length ($F = 25.45$, $p = 0.0007$) and ISI ($F = 5.64$, $p = 0.0415$), as well as their interaction ($F = 10.32$, $p = 0.0106$). Accuracy was lower at the shorter ISI (59.9% at 0.3 s) than at the longer ISI (63.4% at 0.5 s). In addition, performance was substantially higher for two-vibration sequences (72.4%) than for three-vibration sequences (50.9%). For two-vibration sequences, the identification accuracy was largely stable across the ISIs (73.1% at 0.3 s vs. 71.7% at 0.5 s), whereas for three-vibration sequences, the performance improved at the longer interval (46.8% vs. 55.1%). These results indicate that temporal separation becomes increasingly important for longer sequences. Reducing the spatial density from twelve to nine tactors also improved the overall performance relative to Exp. 5, implying that spatial proximity magnifies interference under sequential stimulation (Fig. 11a).

Information-transfer results showed a similar tendency (Fig. 11b). For two-vibration sequences, IT_{est} was 5.14 bits at 0.3 s and 5.20 bits at 0.5 s ($IT_{\mathrm{max}} = 6.34$ bits). For three-vibration sequences, IT_{est} increased to 5.89 bits and 6.50 bits at the respective intervals ($IT_{\mathrm{max}} = 7.76$ bits). Although the overall accuracy decreased for the longer sequences, the higher IT_{est} reflects the enhanced communication capacity and the concentration of confusions among a limited subset of similar sequences.

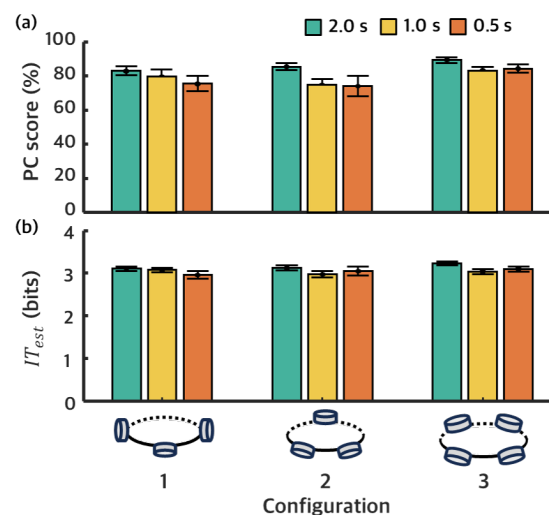

Fig. 12. Identification performance as a function of stimulus duration, grouped by spatial configuration.

C. Discussion

1) Effects of Stimulus Duration: In Exp. 4, vibration duration (1.0 s vs. 0.5 s) showed no significant main effect on identification accuracy. Compared with the 2-s baseline in Exp. 3, however, shorter durations led to modest within-layout decreases (Fig. 12); the PC score dropped by 4.0–11.5% and IT_{est} by 0.19 bits. These results indicate that longer

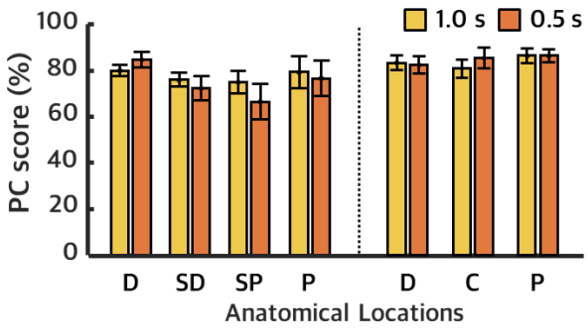

Fig. 13. Percent-correct (PC) scores measured in Exp. 4 across the anatomical locations of forearm for the two vibration durations.

vibrations yielded slightly higher accuracy for single-stimulus localization, but the difference was small.

However, examining the identification performance across anatomical locations suggested that the effect of duration depended on spatial configuration (Fig. 13). When 12 tactors were distributed over the four arm bands, reducing the duration from 1.0 s to 0.5 s led to a considerable decrease in accuracy at the intermediate regions of the forearm, most notably around the sub-proximal band (74.7% to 66.3%). This is in contrast to the identification accuracy at the distal band that increased (79.7% to 84.3%) with the duration decrease. Configurations using three arm bands exhibited a more stable pattern. Apart from a small decrease at the distal band, the performance was largely preserved across locations. The central band accuracy even improved at the shorter duration (80.5% to 85.0%). These different patterns suggest that shortening vibration duration does not always degrade perceptual performance, but alters perceptual load across anatomical locations in a manner that depends on spatial layout. In particular, layouts with fewer arm bands appear more robust to shorter durations, maintaining consistent identification across the forearm.

Although the effect of duration was subtle for single stimuli, it becomes more consequential under sequential presentation, where slight accuracy losses at shorter durations can be compensated by improved temporal efficiency. Information transmission efficiency can be expressed as the information transmission rate in bits per second, obtained by dividing IT_{est} by the stimulus presentation time. Reducing the stimulus duration from 2 s to 0.5 s theoretically increases this rate four-fold, although the actual gain depends on the distinctiveness of the stimuli and the level of interference between them. Thus, even with the observed IT decrease of 0.13 bits at 0.5 s, shorter vibrations are expected to deliver higher overall throughput, particularly when well-separated vibration pairs are used. Therefore, 0.5 s is an adequate duration for sequential vibrotactile presentation, balancing temporal efficiency and perceptual reliability.

2) Effects of Inter-Stimulus Interval and Sequence Length: We evaluated the effects of temporal spacing in Exp. 5 and 6. For the two-vibration sequences at the 12 tactor configurations, increasing ISI from 0 to 0.5 s improved the full-sequence identification accuracy from 44.0% to 50.6%, but the 1.0-s ISI showed a negligible change. IT_{est} also showed a small gain (4.64 $\rightarrow$ 4.87 $\rightarrow$ 4.95 bits; Fig. 11a and b). This improvement was primarily attributable to the enhanced identification of the second vibration. The PC score for the second vibration increased from 54.9% at 0 s to 61.7% and 61.9% at 0.5 s and 1.0 s, respectively, whereas the PC score for the first vibration showed smaller increases (from 71.9% to 74.9% and 75.6%). These results suggest that increasing temporal spacing preferentially benefits later stimuli in a sequence, which are more vulnerable to interference from preceding vibrations.

The interaction between ISI and sequence length became more pronounced in Exp. 6. With nine tactors, two-vibration sequences showed relatively stable performance across ISIs. The identification accuracy was 73.1% at 0.3 s and 71.7% at 0.5 s, with comparable information transfer values of 5.14 and 5.20 bits. In contrast, three-vibration sequences showed clear improvement at the longer interval. The full-sequence accuracy increased from 46.8% at 0.3 s to 55.1% at 0.5 s, accompanied by an increase in IT_{est} from 5.89 to 6.50 bits. Hence, it seems that temporal separation becomes increasingly important as sequence length increases, reflecting cumulative interference and memory demands.

Sequence length affected the identification performance beyond the effect of temporal spacing. Under the matched spatial layout and duration used in Exp. 5 (four tactors per band across three arm bands, 0.5 s), the first vibration in two-vibration sequences was identified less accurately than a single vibration stimulus in Exp. 4. While the single-stimulus accuracy reached 83.0%, the first-stimulus accuracy in two-vibration sequences ranged from 71.9% to 75.6% across the ISIs. This reduction suggests that sequential presentation imposes additional memory and ordering demands, even for the earliest stimulus in the sequence. This burden increased as the interval shortened.

Identification accuracy for sequential stimuli also varied systematically with anatomical location (Fig. 14). In Exp. 5, using the 12-tactor configuration, two-vibration sequences showed the lowest accuracy at distal locations, followed by central and proximal regions, with mean PC scores of 58.9%, 65.0%, and 76.6%, respectively. This ordering was consistent for both the first and second stimuli and remained largely unchanged across ISIs.

Exp. 6 with nine tactors demonstrated consistent patterns with those in Exp. 5. The full-sequence identification accuracy was higher for two-vibration sequences than three-vibration sequences when averaged across ISIs (72.41% vs. 50.93%). The first-vibration identification accuracy also decreased as the sequence length increased. At 0.5 s of ISI, the accuracy declined from 86.3% with two vibrations to 84.3% with three, and at 0.3 s, from 89.6% to 80.0%.

In Exp. 6, the pattern identification performance under sequential stimulation was also affected the anatomical locations of stimulation. For two-vibration sequences, the accuracy increased progressively from distal to proximal locations, with PC scores of 79.7%, 83.1%, and 89.2%, respectively.

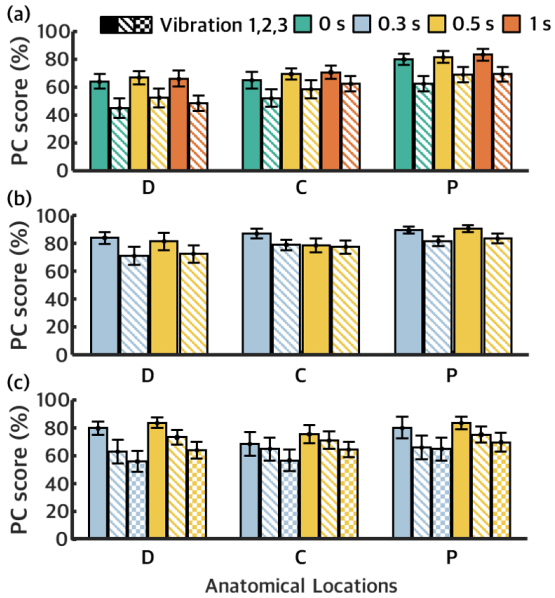

Fig. 14. Identification accuracy for sequential vibrotactile stimuli as a function of anatomical location and inter-stimulus interval. (a) shows results for two-vibration sequences using the 12-tactor configuration in Exp. 5. (b) and (c) show results for two- and three-vibration sequences, respectively, using the 9-tactor configuration in Exp. 6.

This spatial trend is consistent with the pattern observed in Exp. 5. For three-vibration sequences, however, a different spatial profile emerged. The identification accuracy at the distal band reached 71.7%, exceeding that at the central band (68.5%), while the proximal band remained the most reliable location with an accuracy of 75.1%. Notably, when the first vibration was delivered at the distal band, its identification accuracy remained high even in three-vibration sequences (83.8%). Across all tested conditions, the proximal region consistently yielded the highest accuracy regardless of the stimulus positions within the sequence. These observations indicate that the cost of increasing sequence length is not uniformly distributed across the forearm. Instead, identification performance depends jointly on anatomical location and stimulus order, highlighting the importance of considering spatial placement when designing sequential vibrotactile patterns.

Taken all together, these findings suggest that reliable sequential vibrotactile perception on the forearm cannot be achieved by tuning temporal parameters alone. Instead, effective design requires jointly considering array density, sequence length, and anatomical placement, as their interactions determine where and when interference emerges.

VI. General Discussion

A. Summary of Main Findings

This study systematically examined how spatial and temporal parameters constrain vibrotactile information transfer on the forearm. The first set of experiments (Exp. 1–3) identified the spatial limits of localization. Performance remained stable up to four tactors per arm band and declined sharply beyond this density, whereas using three to four such arm bands improved accuracy from 85% to 93% (IT_{est} = 3.1–3.2 bits).

The second set of experiments (Exp. 4–6) revealed temporal characteristics. Vibration duration showed no significant main effect on single-vibration identification. However, duration-dependent changes emerged across anatomical locations. Shorter stimuli (0.5 s) disproportionately reduced accuracy at intermediate forearm regions when four arm bands were used, whereas configurations with three bands maintained more uniform performance. The effect of duration depended on spatial layout rather than acting uniformly across the forearm.

For sequential stimulation, moderate ISIs around 0.5 s effectively mitigated interference between successive vibrations, with the benefit becoming more pronounced as sequence length increased. While two-vibration sequences were relatively robust to ISI, three-vibration sequences showed substantial degradation at shorter intervals, highlighting cumulative temporal interference and memory demands. Reducing spatial density from twelve to nine tactors further improved both identification accuracy and information transfer.

Overall, our results delineate the perceptual bandwidth of forearm-mounted vibrotactile displays. Spatial complexity, temporal overlap, sequence length, and anatomical placement jointly constrain tactile identifiability. As such, efficient information delivery requires balancing array density with temporal parameters rather than optimizing either dimension in isolation.

B. Comparison with Prior Work and Paths for Improvement

Previous studies have reported a broad range of information transfer (IT) values depending on stimulation modality, body site, and coding scheme. Tan and Choi [26] summarized that wearable tactile displays on the arm or torso typically deliver 1–4 bits per decision for single site or simple pattern identification, while spatiotemporal coding or multi-site integration can extend this range to 6–7 bits. For instance, a three-finger display achieved 6.5 bits [57], and four-point apparent-motion patterns on a smartphone reached 7.0 bits [58]. Earlier work further estimated that, under highly optimized conditions, haptic communication can attain information rates of 5–7 bits/s and potentially up to 10–15 bit/s when temporal and intensity dimensions are jointly encoded [55].

The present study achieved IT_{est} values of 3.1–3.2 bits for single-vibration localization and 5.2–6.5 bits for two- and three-vibration sequences on the forearm using uniform vibration intensity. These results are comparable to, or slightly below, those of the multi-finger or multi-site tactile systems that leverage cross-body or motion-based cues. It is noted that our study used a forearm-mounted wearable array that prioritizes spatial compactness and mechanical simplicity; systems for hands, although more sensitive, generally require higher complexity in design.

Further improvement in information transmission efficiency will likely depend on optimizing both the spatiotemporal structure and cognitive mapping of stimuli. The current task

required participants to memorize arbitrary vibration locations, which may have imposed additional working memory load. Encoding meaningful associations, such as mapping vibration patterns to linguistic or symbolic units, could enable users to interpret spatial and temporal variations more naturally and effectively. Such semantic or hierarchical coding is expected to reduce cognitive overhead and increase information throughput beyond the levels achievable through simple spatial memorization. This approach offers a promising direction for transforming forearm-mounted tactile arrays from low-level localization interfaces into high-capacity communication channels capable of conveying structured information.

C. Limitations and Future Work

The present experiments sampled a subset of possible spatial and temporal configurations. Although the tested layouts in Sec. IV covered a wide range of spatial layouts and band counts, additional configurations with different band arrangements or inter-band spacing may reveal finer spatial dependencies that were not observed in this study. Besides, temporal parameters were probed at a few values (duration 0.5 and 1.0 s; ISI 0, 0.3, 0.5, and 1.0 s). Finer sampling around these points and testing nonuniform or asymmetric ISIs within a sequence may locate the onset of interference more precisely and improve sequential identifiability. Varying the ISI between the second and third vibrations, or adapting ISI to the spatial separation of successive locations, are practical next steps.

All stimuli were presented with the same vibration intensity and frequency to isolate spatiotemporal effects. This simplification provided experimental control but does not fully reflect the richness of real tactile communication. Future work will incorporate graded amplitude or frequency modulation to explore multi-dimensional encoding strategies and evaluate how additional stimulus features contribute to overall information transmission capacity.

Finally, the current task relied solely on spatial identification without semantic or linguistic association. While this design allowed precise quantification of perceptual limits, it does not represent how people interpret tactile information in communication contexts. Subsequent studies will investigate whether meaningful tactile–linguistic mappings can facilitate learning, reduce cognitive load, and ultimately increase transmission efficiency, focusing on the authors' native language (Korean).

These directions build on the present results and move toward combining spatial, temporal, and semantic coding for higher-capacity tactile communication.

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